

SANTA YNEZ, CA, USA

WMO: 723762

Lat: 34.607N

Lon: 120.076W

Elev: 671

StdP: 14.34

Time Zone: -8.00 (NAP)

Period: 09-19

WBAN: 93244

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	31.5	32.3	16.3	12.8	60.9	19.4	14.9	60.1	20.4	56.5	18.2	56.2	1.7	70	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.7	96.6	65.9	91.3	64.7	89.6	64.6	68.7	88.9	67.1	87.4	65.9	85.9	10.7	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	82.6	76.3	59.2	76.8	73.5	57.3	71.7	70.9	33.1	88.7	31.9	86.9	30.8	85.3	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.9	17.0	15.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB	24.3	71.0	2.6	2.1	22.4	72.5	20.9	73.7	19.4	74.8	17.5	76.3
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	60.4	52.2	52.2	55.3	58.7	61.0	65.5	69.4	69.9	69.1	63.9	56.2	51.2	(6)
(7)		DBStd	8.35	4.91	5.71	5.15	4.59	4.17	4.54	3.91	3.77	5.24	5.36	5.57	5.28	(7)
(8)		HDD50	156	35	41	12	2	0	0	0	0	0	0	14	52	(8)
(9)		HDD65	2240	397	359	305	194	138	43	5	2	14	86	269	428	(9)
(10)		CDD50	3963	104	102	175	264	341	466	602	616	573	431	200	89	(10)
(11)		CDD65	567	0	0	3	5	14	59	141	153	136	51	5	0	(11)
(12)		CDH74	9463	48	84	178	355	504	1160	1918	1971	1871	1057	296	21	(12)
(13)		CDH80	4004	4	8	45	94	188	471	837	905	907	456	88	1	(13)
(14)	Wind	WSAvg	4.9	4.0	4.8	5.1	5.9	6.2	5.8	5.3	4.9	4.5	4.1	3.9	4.2	(14)
(15)	Precipitation	PrecAvg	17.2	3.7	3.8	2.8	1.2	0.4	0.1	0.0	0.1	0.8	1.3	2.9	(15)	
(16)		PrecMax	33.5	16.4	15.9	12.8	4.5	2.6	0.7	0.3	0.1	0.5	4.2	4.4	9.4	(16)
(17)		PrecMin	6.8	0.2	0.2	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.3	(17)
(18)		PrecStd	6.7	3.6	3.3	2.8	1.1	0.6	0.2	0.1	0.0	0.1	0.9	1.1	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.4	81.2	86.1	88.1	91.0	97.5	97.2	99.1	100.1	97.4	88.4	77.2	(19)
(20)			MCWB	52.5	56.4	58.9	59.3	60.2	65.9	67.3	67.0	65.6	61.8	58.4	54.1	(20)
(21)		2%	DB	73.2	76.9	79.1	82.0	85.9	90.7	91.4	94.6	95.1	90.5	81.7	71.8	(21)
(22)			MCWB	53.0	55.6	57.4	58.4	60.9	65.3	66.4	66.7	65.7	61.4	56.1	53.2	(22)
(23)		5%	DB	70.2	71.7	73.1	78.6	79.3	85.5	89.6	89.8	90.2	85.8	76.9	66.2	(23)
(24)			MCWB	52.8	54.4	56.2	57.4	59.5	63.8	66.4	66.5	64.7	59.7	54.3	53.6	(24)
(25)		10%	DB	64.2	64.2	69.5	72.8	73.2	81.0	85.0	85.6	85.4	80.5	71.7	62.8	(25)
(26)			MCWB	51.7	53.0	55.6	56.6	58.2	62.3	65.0	65.2	63.9	58.3	53.4	51.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.8	60.4	63.8	64.8	64.4	68.4	69.7	70.6	71.1	67.2	60.5	60.0	(27)
(28)			MCD B	70.0	68.3	76.9	79.8	84.9	91.2	90.7	92.1	91.7	82.0	77.6	68.8	(28)
(29)		2%	WB	56.5	58.4	60.3	61.3	62.1	66.2	68.1	68.1	68.1	64.4	58.7	57.9	(29)
(30)			MCD B	65.7	68.4	72.3	75.0	80.5	86.8	89.1	89.0	87.2	82.2	73.6	64.5	(30)
(31)		5%	WB	54.8	56.5	58.2	59.5	60.3	64.4	66.6	66.5	66.1	62.4	57.1	55.6	(31)
(32)			MCD B	64.0	68.0	69.9	73.6	76.9	83.7	86.8	87.6	86.0	78.7	70.4	61.6	(32)
(33)		10%	WB	53.2	54.3	56.3	57.7	58.7	62.7	65.0	65.0	64.4	60.4	55.3	53.7	(33)
(34)			MCD B	60.8	64.0	66.8	70.4	73.7	80.2	83.9	85.1	83.0	75.3	67.3	59.3	(34)
(35)	Mean Daily Temperature Range		MDBR	26.4	25.7	26.0	28.6	26.8	30.0	31.7	32.2	32.5	31.8	30.3	24.5	(35)
(36)		5% DB	MCD BR	35.8	36.9	35.7	38.0	36.8	39.6	38.0	39.3	39.4	41.8	41.0	31.1	(36)
(37)			MCWBR	18.9	19.2	18.0	17.3	16.2	15.8	14.9	14.7	15.0	16.9	18.8	16.6	(37)
(38)		5% WB	MCD BR	26.9	29.6	29.2	30.7	31.8	34.9	35.4	35.7	33.8	32.3	32.9	21.8	(38)
(39)			MCWBR	16.2	16.8	16.6	16.0	15.0	15.2	14.3	14.4	13.9	14.1	16.8	13.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.295	0.309	0.315	0.346	0.360	0.348	0.372	0.353	0.349	0.337	0.298	0.300	(40)	
(41)		taud	2.603	2.577	2.563	2.451	2.433	2.470	2.451	2.473	2.492	2.524	2.613	2.601	(41)	
(42)		Ebn at Noon	293	300	305	298	293	295	287	292	289	285	290	284	(42)	
(43)		Edh at Noon	24	27	30	36	36	35	36	34	32	29	23	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	961	1238	1681	2099	2331	2426	2327	2167	1857	1424	1086	867	(44)	
(45)		RadStd	75	129	108	121	146	108	84	82	79	101	58	72	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	+1.08	N/A	-2.61	+1.21	N/A	N/A	N/A	-230	+392	+148

Nomenclature: See separate page